DATA ANALYTICS STAGE 1 TASK

Data Cleaning, Preparation & Title Optimization Report

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1. Introduction

This project focuses on cleaning and preparing a raw product dataset for marketing analysis. The dataset contains product-related information such as product IDs, titles, descriptions, bullet points, product type id, and product length.

The primary objective of this task is to improve data quality by addressing issues such as missing values, duplicates, and inconsistent formatting. In addition, a new feature called **short_title** was created to enhance search engine optimization (SEO) and improve product readability by generating concise and meaningful product titles.

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2. Dataset Overview

The dataset consists of the following columns:

- **product_id** – Unique identifier for each product
- **title** – Full product title
- **bullet_points** – Key product features
- **description** – Product description
- **product_type_id** – Product category identifier
- **product_length** – Numeric attribute of the product

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3. Data Profiling and Issues Identified

A data profiling process was conducted to assess the quality and structure of the dataset. The following issues were identified:

3.1 Missing Values

- **bullet_points**: 41% missing values
- **description**: 56% missing values
- **product_type_id**: 5% missing values

- **product_length**: 5% missing values

3.2 Duplicate Records

- A total of **306 duplicate entries** were identified in the **product_id** column, which should be unique.

3.3 Data Inconsistencies

- Product titles were excessively long and contained unnecessary words.
- Presence of extra spaces and inconsistent formatting in text fields.
- Use of extra phrases such as “Set of”, “includes”, and “features”.

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4. Data Cleaning Process

To improve the dataset quality, the following steps were performed using Microsoft Excel:

4.1 Removal of Duplicates

Duplicate records were removed based on the **product_id** column using Excel’s *Remove Duplicates* feature. This ensured that each product entry is unique and reliable.

4.2 Handling Missing Values

Different strategies were applied depending on the nature of each column:

- **bullet_points**: Missing values were replaced with *“Not Available”*
- **description**: Missing values were replaced with *“No Description”*

4.3 Text Cleaning and Standardization

To ensure consistency and readability:

- Extra spaces were removed using the TRIM function
- Unnecessary words and symbols were removed using SUBSTITUTE
- Column names were standardized to lowercase with underscores (e.g., product_id)

4.4 Data Validation

The dataset was reviewed to ensure:

- No remaining missing values
- No duplicate product IDs
- No invalid or inconsistent entries

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5. Short Title Creation

5.1 Objective

The goal of the **short_title** feature is to create concise, clear, and SEO-friendly product titles that retain essential information while improving readability.

5.2 Methodology

The following approach was used:

1. Removal of unnecessary words such as “Set of”, “Pack of”, and “Combo of”
2. Trimming extra spaces
3. Limiting title length to approximately 40 characters
4. Extracting key components:
    - Product name
    - Product type
    - Important attributes (e.g., quantity)

An Excel formula was used as a base:

=LEFT(TRIM([@title]),40)

Additional logic was applied to append quantity information (e.g., “2 PCS”) where available.

5.3 Examples

Original Title	Short Title
ArtzFolio Tulip Flowers Blackout Curtain for Door, Window & Room	Tulip Blackout Curtain - 2 PCS
Marks & Spencer Girls' Pyjama Sets Navy Mix 9-10Y	Girls Navy Pyjama Set - 9-10Y
Stainless Steel Water Bottle with Leak Proof Cap 1L	Steel Water Bottle - 1L

5.4 Final Optimization

After applying formulas, manual refinement was performed on selected entries to ensure:

- Clarity and readability
- Inclusion of key product details
- Consistency in formatting

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6. Results and Improvements

After the data cleaning and transformation process:

- All duplicate records were successfully removed
- Missing values were handled appropriately
- Text fields were standardized and cleaned
- Product titles were optimized for readability and SEO
- A new **short_title** column was successfully created

The dataset is now:

- Clean and consistent
- Ready for analysis
- Suitable for marketing and SEO applications

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7. Conclusion

The data cleaning and preparation process significantly improved the quality and usability of the dataset. By addressing missing values, removing duplicates, and standardizing formats, the dataset is now reliable for further analysis.

The introduction of the **short_title** feature enhances product presentation, making it more concise and effective for marketing purposes. This ensures better user engagement and improved search visibility.

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8. Deliverables

The following deliverables have been prepared:

- Cleaned dataset with **short_title** column
- Data profiling summary
- Technical report documenting the entire process